**Solution**

To prove the given points are collinear, it suffices to prove $|A| = \begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix} = 0$ .

Applying $C_1 \to C_1 + C_2$, we deduce that

$$|A| = \begin{vmatrix} a+b+c & b+c & 1 \\ a+b+c & c+a & 1 \\ a+b+c & a+b & 1 \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & b+c & 1 \\ 1 & c+a & 1 \\ 1 & a+b & 1 \end{vmatrix} = (a+b+c)\times 0 = 0$$

which shows that the given points are collinear.

## 7.2.11 Singular and non-singular Matrices

**Definition 7.21**

A square matrix $A$ is said to be **singular** if $|A| = 0$. A square matrix $A$ is said to be **non-singular** if $|A| \neq 0$.

For instance, the matrix $A = \begin{bmatrix} 3 & 8 & 1 \\ -4 & 1 & 1 \\ -4 & 1 & 1 \end{bmatrix}$ is a singular matrix, since

$$|A| = 3(1-1) - 8(-4+4) + 1(-4+4) = 0.$$

If $B = \begin{bmatrix} 2 & 6 & 1 \\ -3 & 0 & 5 \\ 5 & 4 & -7 \end{bmatrix}$ then $|B| = 2(0 - 20) - (-3)(-42 - 4) + 5(30 - 0) = -28 \neq 0$.

Thus $B$ is a non-singular matrix.

**Note 7.14**

If $A$ and $B$ are non-singular matrices of the same order then $AB$ and $BA$ are also non-singular matrices because $|AB| = |A|\,|B| = |BA|$.

## EXERCISE 7.4

(1) Find the area of the triangle whose vertices are (0, 0), (1, 2) and (4, 3).

(2) If $(k, 2)$, (2, 4) and (3, 2) are vertices of the triangle of area 4 square units then determine the value of $k$.

(3) Identify the singular and non-singular matrices:

(i) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{bmatrix}$ (iii) $\begin{bmatrix} 0 & a-b & k \\ b-a & 0 & 5 \\ -k & -5 & 0 \end{bmatrix}$



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